

Sehrish Siddique

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GitHub | LinkedIn | Portfolio

EDUCATION

Quaid-i-Azam University

Bachelor of Science in Computer Science

Expected Completion: Sep 2026

Sep 2022 – Present

Islamabad, Pakistan

CGPA: 3.25/4.00

Relevant Coursework: Software Construction, Data Structure, Operating System, Analysis and Design of Algorithms, Computer Organization & Assembly Language, Object Oriented Programming, Database Systems, Compiler Construction, Computer Communication & Networks.

PROJECTS

Internet of Underwater Things (IoUT) Authentication

Feb 2026 – Jun 2026

Research Project | Learning-Based Physical-Layer Authentication

Project

- Collaborated in a 2-member team to conduct a comparative review of 10 research studies on physical-layer authentication for defending underwater acoustic networks against spoofing and impersonation attacks.
- Developed a multidimensional taxonomy comparing authentication features, learning and decision mechanisms, mobility support, resource requirements, and deployment conditions.
- Evaluated trade-offs across SVM, reinforcement learning, RNN/LSTM, CNN, cooperative, time-reversal, and database-based approaches, identifying research gaps in lightweight authentication, mobility, benchmark datasets, adaptive methods, and practical deployment.

FoodBridge – Local Food Distribution System

Nov 2025 – Present

Final-Year Project | React.js, Node.js, Express.js, MongoDB

Live Demo | GitHub

- Designed and developed a full-stack food donation coordination platform connecting donors, verified NGOs, and administrators across donation posting, request, approval, allocation, and collection workflows.
- Structured the Node.js/Express backend using modular controllers, middleware, models, routes, and services, with role-specific REST APIs and MongoDB/Mongoose data models.
- Implemented JWT-based authentication with HTTP-only cookies, email OTP verification, password hashing, NGO approval controls, and role-based authorization for donor, NGO, and administrator operations.
- Built role-specific React dashboards and donation lifecycle management, including document and image uploads through Cloudinary, notifications, request tracking, and controlled workflow states.

PyCee Compiler

Sep 2025 – Feb 2026

Team Academic Project C++ | Compiler Construction

GitHub

- Collaborated in a 2-member team to design the PyCee programming language and develop its lexical analyser, including tokenization, token classification, and lexical-error detection.
- Independently designed the language grammar and implemented FIRST/FOLLOW set computation and LL(1) parsing-table construction for predictive syntax analysis.
- Applied the resulting compiler front end to language constructs including declarations, expressions, arrays, functions, conditionals, loops, return statements, and function calls.

EXPERIENCE

Techfy

Sep 2026 – Present

Backend Developer Intern

Remote

- Developing a student management backend using Python, Flask, Supabase, and PostgreSQL, with RESTful APIs for creating, retrieving, updating, and deleting student records.
- Implementing authenticated API access and Swagger/OpenAPI documentation to support protected endpoint access, interactive testing, and validation of API requests and responses.

Techohash Solutions

Aug 2024 – Sep 2024

Web Developer Intern

Rawalpindi, Pakistan

- Implemented responsive website updates using HTML5, CSS3, and JavaScript, translating assigned requirements into functional changes across existing web applications.
- Identified and corrected interface, layout, and responsiveness issues using browser developer tools and iterative testing across different screen sizes.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, C++, SQL

Web Development & Backend: HTML5, CSS3, React.js, Tailwind CSS, Flask, Node.js, Express.js, Mongoose, REST APIs

Databases & Backend Services: PostgreSQL, MongoDB, MySQL, Supabase, Cloudinary

Tools & API Development: Git, GitHub, Postman, Swagger/OpenAPI, Visual Studio Code, Overleaf